GENESIS
IDEATHON

TEAM & DOMAIN

- Team Name – Sirius
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- Selected Domain – #6
- Problem Statement – Cancer treatment often damages healthy cells along with tumor cells, causing unwanted side effects. The key challenge is delivering drugs specifically to the tumor, with precise, controlled and timely release, while minimizing exposure to healthy tissue.



DOMAIN PROBLEM IN OUR WORDS

Smart Tumor-Targeted Micro-Capsule Logic Gate

• What exactly needs to be solved?

The key problem is that most targeted drug carriers can be misled. They release their medicine when they detect a single trigger, such as acidity or a particular enzyme, but inflamed or injured healthy tissue can produce the same signals. As a result, the drug may be released outside the tumor, exposing healthy cells to unnecessary harm. What is needed is a carrier that does not depend on just one signal. It should confirm that several tumorrelated signals are present together before releasing its drug, so that it can tell tumor tissue apart from normal tissue more reliably and deliver treatment only where it is needed.

• Why is it important right now?

Cancer cases are rising worldwide, so more patients are exposed to treatments that harm healthy tissue. Healthy tissues are exposed to toxic drugs, causing severe side effects such as hair loss, nausea, weakened immunity and organ damage. Because of this toxicity, doctors often cannot give the full effective dose. Tumors differ from patient to patient, so one target marker is not always reliable.

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WHO IS AFFECTED?

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Target users, stakeholders, beneficiaries

PRIMARY USERS

Cancer patients receiving drug-based treatment, especially those whose treatment is limited by toxicity or whose tumors are difficult to target using one marker.cancer patients receiving drug-based treatment, especially those whose treatment is limited by toxicity or whose tumors are difficult to target using one marker.

SECONDARY STAKEHOLDERS

Medical professionals utilise targeted microcapsules to drastically improve treatment precision and patient compliance:

- Medical & Interventional Oncologists
- Radiologists and Theranostic Specialists

On a commercial scale, the target users acquiring, researching, and implementing the technology include

- Biopharmaceutical & Biotechnology Companies
- Contract Research Organisations (CROs) & Research Institutes

WHO BENEFITS IF SOLVED

The people who benefit the most are patients suffering from solid tumors and aggressive or hard-to-treat malignancies.

- Patients with High-Prevalence Solid Tumors
- Patients with Deep-Tissue or Inoperable Malignancies
- Chemotherapy-Sensitive Patients Facing High Toxicity
- Resistant or Relapsed Cancer Patients

CURRENT SITUATION & CONTEXT

Existing scenario, background and how things work today

Current situation

The Definitive Pillars of Cancer Treatment

01
- Surgery
- Radiation Therapy
- Traditional Chemotherapy

Advanced Biologic and Systemic Therapies

02
- Immunotherapy: I
- Immune Checkpoint Inhibitors:
- CAR T-Cell Therapy
- Targeted Small-Molecule Therapies
- Hormone Therapy

Non-Invasive and Physical Ablation Techniques

03
- Tumor Treating Fields (TTFields)
- Ablation Therapy

•Backbone of care: systemic chemotherapy is still central for many solid tumors, yet only a small share of the drug reaches the tumor.
•Dose is limited by toxicity to bone marrow, heart, gut and hair follicles, not by what the tumor could tolerate.
•Nanocarriers help partly: Doxil, Abraxane and Onivyde lower toxicity but mostly rely on passive accumulation (the EPR effect), which is inconsistent in humans.

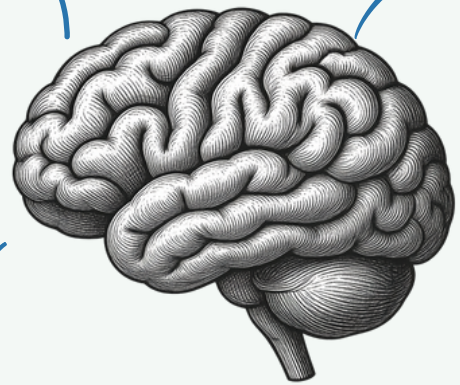
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KEY CHALLENGES

Main difficulties, bottlenecks, root causes and consequences

1. Premature Drug Release and Logic Failure

- Challenge: The micro-capsule may release its payload before reaching the tumor due to leakage, unstable materials, or unintended activation by non-target biological signals. This could reduce targeting accuracy and expose healthy tissues to the drug.
- Risk Mitigation: Improve molecular sensor selectivity and use a multi-input logic gate that requires specific tumor-associated signals before triggering release.
- Performance Evaluation: Measure baseline leakage in the absence of activation signals and specificity under non-target conditions.



2. Biological Barriers and Immune Clearance

- Challenge: Blood proteins may form a coating around the capsule and interfere with its sensing or targeting functions. Additionally, immune cells in organs such as the liver and spleen may remove the capsule before it reaches the tumor.
- Risk Mitigation: Investigate biocompatible materials and suitable surface modifications that may reduce unwanted biological interactions while preserving capsule stability.
- Performance Evaluation: Assess structural stability under simulated circulation conditions, compatibility with healthy cells, and interactions with immune cells.

3. Limited Tumor Penetration and Incomplete Drug Release

- Challenge: Irregular tumor blood vessels and dense surrounding tissue may restrict the capsule's movement into deeper tumor regions. Even after activation, some of the drug may remain trapped inside the capsule.
- Risk Mitigation: Optimize capsule size, material properties, and release mechanisms to improve tissue penetration and payload delivery.
- Performance Evaluation: Measure activation sensitivity under target conditions, release completeness after activation, and the distribution of the released payload within tumor-like tissue.

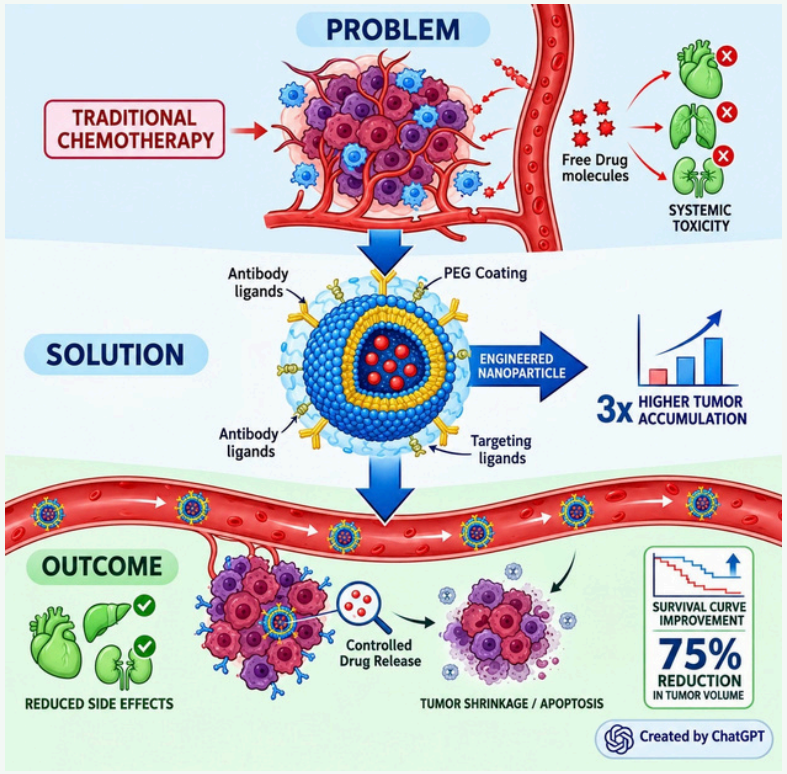
EXISTING SOLUTIONS / APPROACHES

Chemotherapy & Radiation

Can damage healthy tissues along with cancer cells, causing unwanted side effects.
Investigate smart drug delivery that releases its payload in response to specific tumor-related signals, potentially reducing unnecessary exposure to healthy tissues.

Targeted Therapy & Antibody–Drug Conjugates (ADCs)

Targeted drugs may affect non-target tissues, and some drug-linking systems may release their payload before reaching the tumor.
Improve control over when and where the drug is released by using a signal-responsive capsule.



Conventional Targeted Drug Delivery

Reaching the tumor does not guarantee that the drug will penetrate all tumor regions or be released completely.
Evaluate both tumor penetration and controlled drug release to improve delivery reliability.

Existing Smart Drug-Delivery Systems

Biological signals vary across tumor regions, and a capsule may activate incorrectly or fail to activate when required. Its safety after drug release also needs consideration.
Investigate multi-signal logic gates, reliable activation thresholds, and safe capsule breakdown and clearance.

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PROBLEM INSIGHTS & OPPORTUNITY AREAS

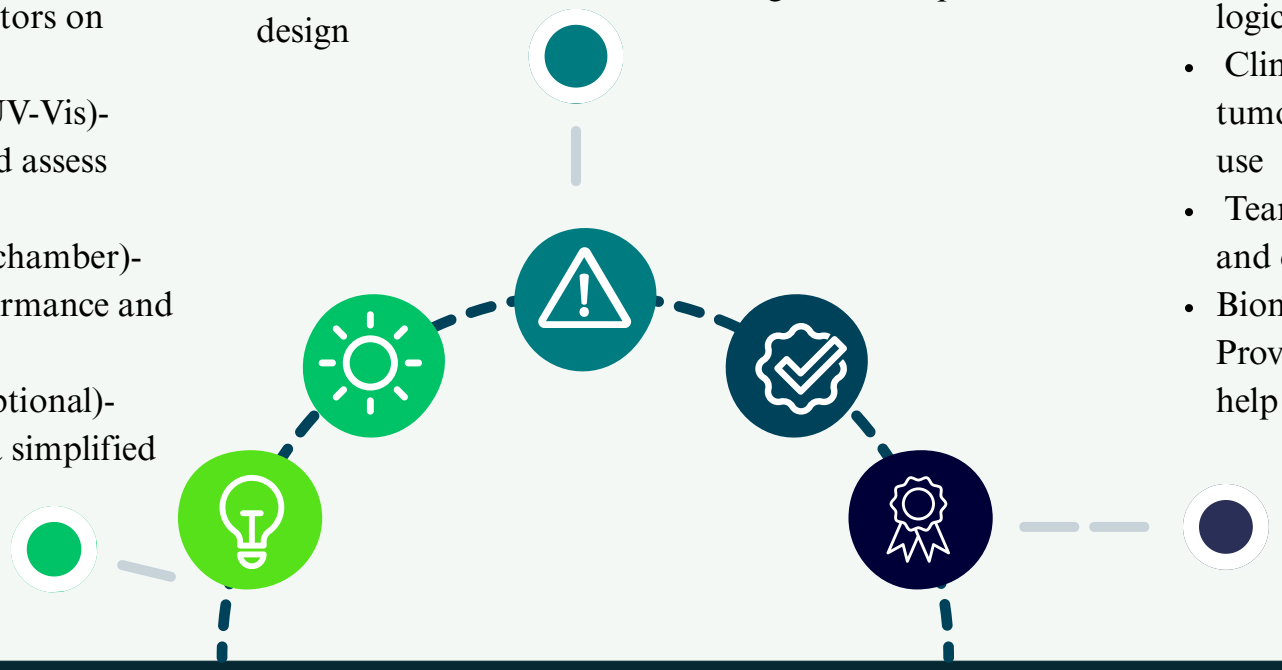
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Component- Purpose

- Microfluidic chip + syringe pump -Produces uniformly sized micro-capsules for consistent performance.
- Biodegradable polymers (PLGA / chitosan)- Form the capsule shell and allow it to break down naturally.
- Smart linkers (pH-, redox-, or enzyme-sensitive)- Act as locks that respond to selected chemical conditions associated with the tumor environment.
- Targeting ligands (folic acid / RGD peptide)- Help the capsule recognize and bind to selected receptors on target cells.
- Characterization tools (DLS, SEM/TEM, UV-Vis)- Measure capsule size, examine structure, and assess drug loading and release.
- Cell culture laboratory (incubator, hypoxia chamber)- Supports controlled testing of capsule performance and effects on cancer and healthy cells.
- Arduino/ESP32 + pH and redox sensors (optional)- Demonstrates the logic-gate concept using a simplified electronic prototype.

Software:

- MATLAB / Python- Models the logic gate: drug is released only if all required signals are present.
- COMSOL Multiphysics -Simulates how the drug spreads and releases inside tumor tissue
- COPASI -Models the chemical reactions that trigger the capsule
- AutoDock Vina / PyMOL-Checks how well the targeting ligand binds to the cancer cell receptor
- ImageJ / GraphPad Prism-Analyzes lab images and results
- BioRender / Draw.io - Creates diagrams to explain the design



Human Resources Required

- Biomedical / Chemical Engineer - Designs and builds the capsule and its smart “locks” Pharmacy / Biology Member-Handles drug loading, release, and cell testing
- Computational Modeller-Simulates the logic gate and drug release
- Clinical Advisor (Oncologist)-Guides tumor targeting and checks real-world use
- Team Lead - Coordinates the project and documentation
- Biomarker Databases (e.g., TCGA): Provide cancer-related molecular data to help identify potential release trigger

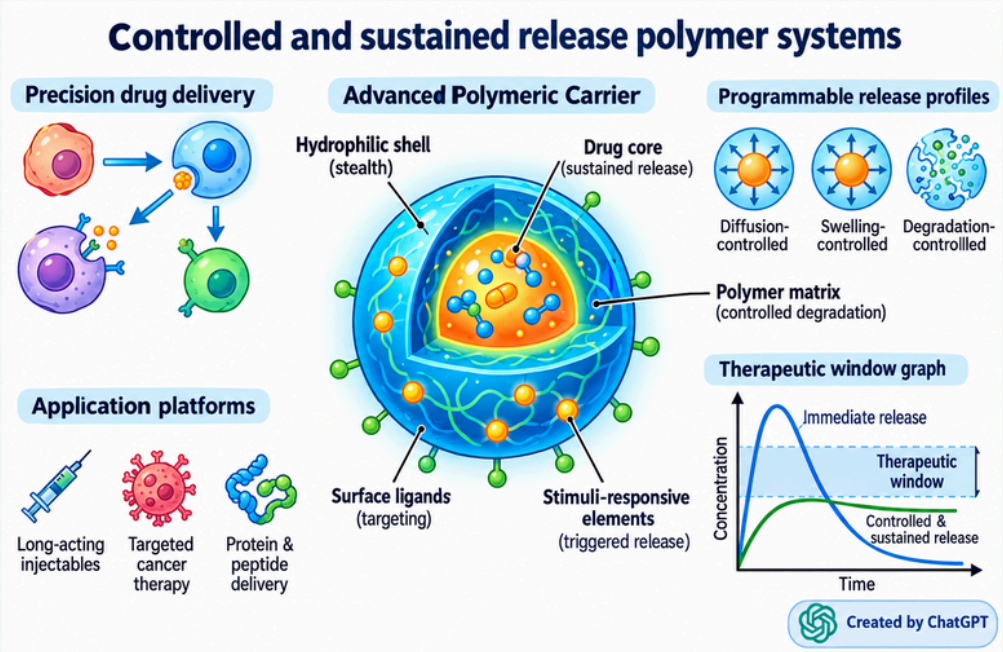
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INITIAL DIRECTION & EXPECTED OUTCOME

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1. Initial Direction — What We Intend to Do

- Design a conceptual drug-delivery micro-capsule that responds to multiple tumor-associated signals.
- Explore an AND logic gate that activates drug release only when two selected signals are detected together.
- Identify suitable tumor-associated signals through scientific literature and research.
- Develop a conceptual model or simulation to demonstrate the proposed mechanism.



2. What We Want to Improve

- Reduce premature drug release in healthy tissues.
- Improve the selectivity of drug release towards the intended tumor environment.
- Reduce unintended exposure of healthy tissues to toxic drugs.
- Address the limitations of relying on a single tumor-associated signal.

3. Expected Outcome — What Successful Improvement Should Achieve

- A clear conceptual design of the multi-signal micro-capsule.
- A working simulation or proof of concept demonstrating AND-gate behaviour.
- Evidence that the proposed system responds more strongly when both selected signals are present than when either signal occurs alone.
- A foundation for future experimental research into more selective drug delivery.